

1000 AI Projects — Foundation to GenAI & Agentic AI

LLM · UI (React / Streamlit) · Backend (Python FastAPI) · AI Frameworks (LangGraph / LangChain / AutoGen / CrewAI / LlamaIndex / DSPy) · Vector DB & Infra

Master in AI Engineering — Build production-grade GenAI, Agentic AI, RAG & MLOps systems.
From LLM foundations to multi-agent orchestration — the complete hands-on path for AI engineers.

Created By Balawant M Kadam
Exported: 20/9/2026, 9:07:51 pm
Projects in export: 60 of 1000
Filters: Level: research | Type: All | Domain: All | Framework: All

#0701

TRADITIONAL

CUTTING-EDGE

Wildfire Prediction Neuro-Symbolic Reasoner

Climate & Environment

Push the frontier with a Neuro-Symbolic Reasoner using PyTorch exploring novel methods for wildfire prediction — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	PyTorch DeepSpeed Axolotl
VECTOR / INFRA	Milvus · FAISS

#0702

TRADITIONAL

CUTTING-EDGE

Biodiversity Tracking Continual Learning Agent

Climate & Environment

Push the frontier with a Continual Learning Agent using DSPy exploring novel methods for biodiversity tracking — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	DSPy PyTorch Weights & Biases Unsloth
VECTOR / INFRA	pgvector · ChromaDB

#0703

TRADITIONAL

CUTTING-EDGE

Carbon Accounting Causal Discovery Engine

Climate & Environment

Push the frontier with a Causal Discovery Engine using LangGraph exploring novel methods for carbon accounting — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM DeepSeek-R1

UI Streamlit + Custom PyTorch viz

BACKEND Python · FastAPI + Distributed tracing (OTel)

AI FRAMEWORKS [LangGraph](#) [vLLM](#) [Triton Inference Server](#)

VECTOR / INFRA Neo4j · Pinecone

#0704

TRADITIONAL

CUTTING-EDGE

Weather Forecasting Alignment Sandbox

Climate & Environment

Push the frontier with a Alignment Sandbox using Ray exploring novel methods for weather forecasting — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Claude 3.7 Sonnet (extended thinking)

UI React + D3 / Observable

BACKEND Python · FastAPI + Experiment orchestration

AI FRAMEWORKS [Ray](#) [LangGraph](#) [PEFT / LoRA](#) [OpenTelemetry](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0705

TRADITIONAL

CUTTING-EDGE

Ocean Data Analytics Emergent Behavior Testbed

Climate & Environment

Push the frontier with a Emergent Behavior Testbed using PyTorch exploring novel methods for ocean data analytics — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Gemini 2.5 Pro

UI React + Three.js + WebGL

BACKEND Python · FastAPI + HPC scheduler integration

AI FRAMEWORKS [PyTorch](#) [TRL](#) [NeMo Guardrails](#)

VECTOR / INFRA DuckDB · Qdrant

#0706

TRADITIONAL

CUTTING-EDGE

Network Optimization Autonomous Research Lab

Telecommunications

Push the frontier with a Autonomous Research Lab using DSPy exploring novel methods for network optimization — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	DSPy PyTorch Axolotl DeepSpeed
VECTOR / INFRA	LanceDB · Milvus

#0707

TRADITIONAL

CUTTING-EDGE

Churn Prediction Neuro-Symbolic Reasoner

Telecommunications

Push the frontier with a Neuro-Symbolic Reasoner using LangGraph exploring novel methods for churn prediction — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	LangGraph Unsloth Weights & Biases
VECTOR / INFRA	OpenSearch · pgvector

#0708

TRADITIONAL

CUTTING-EDGE

Fault Diagnosis Continual Learning Agent

Telecommunications

Push the frontier with a Continual Learning Agent using Ray exploring novel methods for fault diagnosis — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	Ray LangGraph Triton Inference Server vLLM
VECTOR / INFRA	FAISS · Neo4j

#0709

TRADITIONAL

CUTTING-EDGE

Customer Care Assistant Causal Discovery Engine

Telecommunications

Push the frontier with a Causal Discovery Engine using PyTorch exploring novel methods for customer care assistant — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	PyTorch OpenTelemetry PEFT / LoRA
VECTOR / INFRA	ChromaDB · Elasticsearch

#0710

TRADITIONAL

CUTTING-EDGE

5G Slice Management Alignment Sandbox

Telecommunications

Push the frontier with a Alignment Sandbox using DSPy exploring novel methods for 5g slice management — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	DSPy PyTorch NeMo Guardrails TRL
VECTOR / INFRA	Pinecone · DuckDB

#0711

TRADITIONAL

CUTTING-EDGE

Spectrum Analytics Emergent Behavior Testbed

Telecommunications

Push the frontier with a Emergent Behavior Testbed using LangGraph exploring novel methods for spectrum analytics — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	LangGraph DeepSpeed Axolotl
VECTOR / INFRA	Weaviate · LanceDB

#0712

TRADITIONAL

CUTTING-EDGE

Property Valuation Autonomous Research Lab

Real Estate & Construction

Push the frontier with a Autonomous Research Lab using Ray exploring novel methods for property valuation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	Ray LangGraph Weights & Biases Unsloth
VECTOR / INFRA	Qdrant · OpenSearch

#0713

TRADITIONAL

CUTTING-EDGE

Site Safety Monitoring Neuro-Symbolic Reasoner

Real Estate & Construction

Push the frontier with a Neuro-Symbolic Reasoner using PyTorch exploring novel methods for site safety monitoring — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	PyTorch vLLM Triton Inference Server
VECTOR / INFRA	Milvus · FAISS

#0714

TRADITIONAL

CUTTING-EDGE

Lease Abstraction Continual Learning Agent

Real Estate & Construction

Push the frontier with a Continual Learning Agent using DSPy exploring novel methods for lease abstraction — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	DSPy PyTorch PEFT / LoRA OpenTelemetry
VECTOR / INFRA	pgvector · ChromaDB

#0715

TRADITIONAL

CUTTING-EDGE

BIM Analysis Causal Discovery Engine

Real Estate & Construction

Push the frontier with a Causal Discovery Engine using LangGraph exploring novel methods for bim analysis — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	LangGraph TRL NeMo Guardrails
VECTOR / INFRA	Neo4j · Pinecone

#0716

TRADITIONAL

CUTTING-EDGE

Construction Scheduling Alignment Sandbox

Real Estate & Construction

Push the frontier with a Alignment Sandbox using Ray exploring novel methods for construction scheduling — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	Ray LangGraph Axolotl DeepSpeed
VECTOR / INFRA	Elasticsearch · Weaviate

#0717

TRADITIONAL

CUTTING-EDGE

Tenant Support Emergent Behavior Testbed

Real Estate & Construction

Push the frontier with a Emergent Behavior Testbed using PyTorch exploring novel methods for tenant support — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	PyTorch Unsloth Weights & Biases
VECTOR / INFRA	DuckDB · Qdrant

#0718

TRADITIONAL

CUTTING-EDGE

Itinerary Planning Autonomous Research Lab

Travel & Hospitality

Push the frontier with a Autonomous Research Lab using DSPy exploring novel methods for itinerary planning — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	DSPy PyTorch Triton Inference Server vLLM
VECTOR / INFRA	LanceDB · Milvus

#0719

TRADITIONAL

CUTTING-EDGE

Dynamic Pricing Neuro-Symbolic Reasoner

Travel & Hospitality

Push the frontier with a Neuro-Symbolic Reasoner using LangGraph exploring novel methods for dynamic pricing — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	LangGraph OpenTelemetry PEFT / LoRA
VECTOR / INFRA	OpenSearch · pgvector

#0720

TRADITIONAL

CUTTING-EDGE

Review Analytics Continual Learning Agent

Travel & Hospitality

Push the frontier with a Continual Learning Agent using Ray exploring novel methods for review analytics — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	Ray LangGraph NeMo Guardrails TRL
VECTOR / INFRA	FAISS · Neo4j

#0721

TRADITIONAL

CUTTING-EDGE

Virtual Concierge Causal Discovery Engine

Travel & Hospitality

Push the frontier with a Causal Discovery Engine using PyTorch exploring novel methods for virtual concierge — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM DeepSeek-R1

UI Streamlit + Custom PyTorch viz

BACKEND Python · FastAPI + Distributed tracing (OTel)

AI FRAMEWORKS [PyTorch](#) [DeepSpeed](#) [Axolotl](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0722

TRADITIONAL

CUTTING-EDGE

Demand Forecasting Alignment Sandbox

Travel & Hospitality

Push the frontier with a Alignment Sandbox using DSPy exploring novel methods for demand forecasting — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Claude 3.7 Sonnet (extended thinking)

UI React + D3 / Observable

BACKEND Python · FastAPI + Experiment orchestration

AI FRAMEWORKS [DSPy](#) [PyTorch](#) [Weights & Biases](#) [Unsloth](#)

VECTOR / INFRA Pinecone · DuckDB

#0723

TRADITIONAL

CUTTING-EDGE

Multilingual Support Emergent Behavior Testbed

Travel & Hospitality

Push the frontier with a Emergent Behavior Testbed using LangGraph exploring novel methods for multilingual support — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Gemini 2.5 Pro

UI React + Three.js + WebGL

BACKEND Python · FastAPI + HPC scheduler integration

AI FRAMEWORKS [LangGraph](#) [vLLM](#) [Triton Inference Server](#)

VECTOR / INFRA Weaviate · LanceDB

#0724

TRADITIONAL

CUTTING-EDGE

Claims Automation Autonomous Research Lab

Insurance

Push the frontier with a Autonomous Research Lab using Ray exploring novel methods for claims automation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	Ray LangGraph PEFT / LoRA OpenTelemetry
VECTOR / INFRA	Qdrant · OpenSearch

#0725

TRADITIONAL

CUTTING-EDGE

Risk Assessment Neuro-Symbolic Reasoner

Insurance

Push the frontier with a Neuro-Symbolic Reasoner using PyTorch exploring novel methods for risk assessment — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	PyTorch TRL NeMo Guardrails
VECTOR / INFRA	Milvus · FAISS

#0726

TRADITIONAL

CUTTING-EDGE

Policy Comparison Continual Learning Agent

Insurance

Push the frontier with a Continual Learning Agent using DSPy exploring novel methods for policy comparison — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	DSPy PyTorch Axolotl DeepSpeed
VECTOR / INFRA	pgvector · ChromaDB

#0727

TRADITIONAL

CUTTING-EDGE

Damage Estimation Causal Discovery Engine

Insurance

Push the frontier with a Causal Discovery Engine using LangGraph exploring novel methods for damage estimation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	LangGraph Unsloth Weights & Biases
VECTOR / INFRA	Neo4j · Pinecone

#0728

TRADITIONAL

CUTTING-EDGE

Fraud Detection Alignment Sandbox

Insurance

Push the frontier with a Alignment Sandbox using Ray exploring novel methods for fraud detection — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	Ray LangGraph Triton Inference Server vLLM
VECTOR / INFRA	Elasticsearch · Weaviate

#0729

TRADITIONAL

CUTTING-EDGE

Underwriting Copilot Emergent Behavior Testbed

Insurance

Push the frontier with a Emergent Behavior Testbed using PyTorch exploring novel methods for underwriting copilot — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	PyTorch OpenTelemetry PEFT / LoRA
VECTOR / INFRA	DuckDB · Qdrant

#0730

TRADITIONAL

CUTTING-EDGE

Code Review Autonomous Research Lab

Software Engineering

Push the frontier with a Autonomous Research Lab using DSPy exploring novel methods for code review — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	DSPy PyTorch NeMo Guardrails TRL
VECTOR / INFRA	LanceDB · Milvus

#0731

TRADITIONAL

CUTTING-EDGE

Test Generation Neuro-Symbolic Reasoner

Software Engineering

Push the frontier with a Neuro-Symbolic Reasoner using LangGraph exploring novel methods for test generation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	LangGraph DeepSpeed Axolotl
VECTOR / INFRA	OpenSearch · pgvector

#0732

TRADITIONAL

CUTTING-EDGE

Incident Root Cause Continual Learning Agent

Software Engineering

Push the frontier with a Continual Learning Agent using Ray exploring novel methods for incident root cause — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	Ray LangGraph Weights & Biases Unsloth
VECTOR / INFRA	FAISS · Neo4j

#0733

TRADITIONAL

CUTTING-EDGE

Documentation Causal Discovery Engine

Software Engineering

Push the frontier with a Causal Discovery Engine using PyTorch exploring novel methods for documentation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	PyTorch vLLM Triton Inference Server
VECTOR / INFRA	ChromaDB · Elasticsearch

#0734

TRADITIONAL

CUTTING-EDGE

Legacy Migration Alignment Sandbox

Software Engineering

Push the frontier with a Alignment Sandbox using DSPy exploring novel methods for legacy migration — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	DSPy PyTorch PEFT / LoRA OpenTelemetry
VECTOR / INFRA	Pinecone · DuckDB

#0735

TRADITIONAL

CUTTING-EDGE

Developer Copilot Emergent Behavior Testbed

Software Engineering

Push the frontier with a Emergent Behavior Testbed using LangGraph exploring novel methods for developer copilot — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	LangGraph TRL NeMo Guardrails
VECTOR / INFRA	Weaviate · LanceDB

#0736

TRADITIONAL

CUTTING-EDGE

Clinical Notes Autonomous Research Lab

Healthcare

Push the frontier with a Autonomous Research Lab using Ray exploring novel methods for clinical notes — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	Ray LangGraph Axolotl DeepSpeed
VECTOR / INFRA	Qdrant · OpenSearch

#0737

TRADITIONAL

CUTTING-EDGE

Diagnostic Imaging Neuro-Symbolic Reasoner

Healthcare

Push the frontier with a Neuro-Symbolic Reasoner using PyTorch exploring novel methods for diagnostic imaging — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	PyTorch Unsloth Weights & Biases
VECTOR / INFRA	Milvus · FAISS

#0738

TRADITIONAL

CUTTING-EDGE

Patient Triage Continual Learning Agent

Healthcare

Push the frontier with a Continual Learning Agent using DSPy exploring novel methods for patient triage — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	DSPy PyTorch Triton Inference Server vLLM
VECTOR / INFRA	pgvector · ChromaDB

#0739

TRADITIONAL

CUTTING-EDGE

Drug Interaction Checker Causal Discovery Engine

Healthcare

Push the frontier with a Causal Discovery Engine using LangGraph exploring novel methods for drug interaction checker — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	LangGraph OpenTelemetry PEFT / LoRA
VECTOR / INFRA	Neo4j · Pinecone

#0740

TRADITIONAL

CUTTING-EDGE

EHR Records Alignment Sandbox

Healthcare

Push the frontier with a Alignment Sandbox using Ray exploring novel methods for ehr records — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	Ray LangGraph NeMo Guardrails TRL
VECTOR / INFRA	Elasticsearch · Weaviate

#0741

TRADITIONAL

CUTTING-EDGE

Remote Patient Monitoring Emergent Behavior Testbed

Healthcare

Push the frontier with a Emergent Behavior Testbed using PyTorch exploring novel methods for remote patient monitoring — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	PyTorch DeepSpeed Axolotl
VECTOR / INFRA	DuckDB · Qdrant

#0742

TRADITIONAL

CUTTING-EDGE

Fraud Detection Autonomous Research Lab

Finance & Banking

Push the frontier with a Autonomous Research Lab using DSPy exploring novel methods for fraud detection — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	DSPy PyTorch Weights & Biases Unsloth
VECTOR / INFRA	LanceDB · Milvus

#0743

TRADITIONAL

CUTTING-EDGE

Credit Scoring Neuro-Symbolic Reasoner

Finance & Banking

Push the frontier with a Neuro-Symbolic Reasoner using LangGraph exploring novel methods for credit scoring — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	LangGraph vLLM Triton Inference Server
VECTOR / INFRA	OpenSearch · pgvector

#0744

TRADITIONAL

CUTTING-EDGE

Portfolio Analytics Continual Learning Agent

Finance & Banking

Push the frontier with a Continual Learning Agent using Ray exploring novel methods for portfolio analytics — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	Ray LangGraph PEFT / LoRA OpenTelemetry
VECTOR / INFRA	FAISS · Neo4j

#0745

TRADITIONAL

CUTTING-EDGE

Loan Underwriting Causal Discovery Engine

Finance & Banking

Push the frontier with a Causal Discovery Engine using PyTorch exploring novel methods for loan underwriting — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM DeepSeek-R1

UI Streamlit + Custom PyTorch viz

BACKEND Python · FastAPI + Distributed tracing (OTel)

AI FRAMEWORKS [PyTorch](#) [TRL](#) [NeMo Guardrails](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0746

TRADITIONAL

CUTTING-EDGE

Transaction Reconciliation Alignment Sandbox

Finance & Banking

Push the frontier with a Alignment Sandbox using DSPy exploring novel methods for transaction reconciliation — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Claude 3.7 Sonnet (extended thinking)

UI React + D3 / Observable

BACKEND Python · FastAPI + Experiment orchestration

AI FRAMEWORKS [DSPy](#) [PyTorch](#) [Axolotl](#) [DeepSpeed](#)

VECTOR / INFRA Pinecone · DuckDB

#0747

TRADITIONAL

CUTTING-EDGE

Market Sentiment Emergent Behavior Testbed

Finance & Banking

Push the frontier with a Emergent Behavior Testbed using LangGraph exploring novel methods for market sentiment — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM Gemini 2.5 Pro

UI React + Three.js + WebGL

BACKEND Python · FastAPI + HPC scheduler integration

AI FRAMEWORKS [LangGraph](#) [Unsloth](#) [Weights & Biases](#)

VECTOR / INFRA Weaviate · LanceDB

#0748

TRADITIONAL

CUTTING-EDGE

Product Recommendations Autonomous Research Lab

Retail & E-commerce

Push the frontier with a Autonomous Research Lab using Ray exploring novel methods for product recommendations — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	Ray LangGraph Triton Inference Server vLLM
VECTOR / INFRA	Qdrant · OpenSearch

#0749

TRADITIONAL

CUTTING-EDGE

Demand Forecasting Neuro-Symbolic Reasoner

Retail & E-commerce

Push the frontier with a Neuro-Symbolic Reasoner using PyTorch exploring novel methods for demand forecasting — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	PyTorch OpenTelemetry PEFT / LoRA
VECTOR / INFRA	Milvus · FAISS

#0750

TRADITIONAL

CUTTING-EDGE

Visual Search Continual Learning Agent

Retail & E-commerce

Push the frontier with a Continual Learning Agent using DSPy exploring novel methods for visual search — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	DSPy PyTorch NeMo Guardrails TRL
VECTOR / INFRA	pgvector · ChromaDB

#0751

TRADITIONAL

CUTTING-EDGE

Dynamic Pricing Causal Discovery Engine

Retail & E-commerce

Push the frontier with a Causal Discovery Engine using LangGraph exploring novel methods for dynamic pricing — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	LangGraph DeepSpeed Axolotl
VECTOR / INFRA	Neo4j · Pinecone

#0752

TRADITIONAL

CUTTING-EDGE

Cart Abandonment Alignment Sandbox

Retail & E-commerce

Push the frontier with a Alignment Sandbox using Ray exploring novel methods for cart abandonment — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	Ray LangGraph Weights & Biases Unsloth
VECTOR / INFRA	Elasticsearch · Weaviate

#0753

TRADITIONAL

CUTTING-EDGE

Inventory Optimization Emergent Behavior Testbed

Retail & E-commerce

Push the frontier with a Emergent Behavior Testbed using PyTorch exploring novel methods for inventory optimization — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	PyTorch vLLM Triton Inference Server
VECTOR / INFRA	DuckDB · Qdrant

#0754

TRADITIONAL

CUTTING-EDGE

Adaptive Learning Paths Autonomous Research Lab

Education

Push the frontier with a Autonomous Research Lab using DSPy exploring novel methods for adaptive learning paths — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	DSPy PyTorch PEFT / LoRA OpenTelemetry
VECTOR / INFRA	LanceDB · Milvus

#0755

TRADITIONAL

CUTTING-EDGE

Essay Grading Neuro-Symbolic Reasoner

Education

Push the frontier with a Neuro-Symbolic Reasoner using LangGraph exploring novel methods for essay grading — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Open-weight MoE (custom)
UI	Next.js + Jupyter embed
BACKEND	Python · FastAPI + DeepSpeed + NCCL
AI FRAMEWORKS	LangGraph TRL NeMo Guardrails
VECTOR / INFRA	OpenSearch · pgvector

#0756

TRADITIONAL

CUTTING-EDGE

Tutoring Assistant Continual Learning Agent

Education

Push the frontier with a Continual Learning Agent using Ray exploring novel methods for tutoring assistant — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	o3 / o4-mini (reasoning)
UI	React + Live experiment tracker
BACKEND	Python · FastAPI + Custom CUDA workers
AI FRAMEWORKS	Ray LangGraph Axolotl DeepSpeed
VECTOR / INFRA	FAISS · Neo4j

#0757

TRADITIONAL

CUTTING-EDGE

Curriculum Generator Causal Discovery Engine

Education

Push the frontier with a Causal Discovery Engine using PyTorch exploring novel methods for curriculum generator — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	DeepSeek-R1
UI	Streamlit + Custom PyTorch viz
BACKEND	Python · FastAPI + Distributed tracing (OTel)
AI FRAMEWORKS	PyTorch Unsloth Weights & Biases
VECTOR / INFRA	ChromaDB · Elasticsearch

#0758

TRADITIONAL

CUTTING-EDGE

Student Engagement Analytics Alignment Sandbox

Education

Push the frontier with a Alignment Sandbox using DSPy exploring novel methods for student engagement analytics — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Claude 3.7 Sonnet (extended thinking)
UI	React + D3 / Observable
BACKEND	Python · FastAPI + Experiment orchestration
AI FRAMEWORKS	DSPy PyTorch Triton Inference Server vLLM
VECTOR / INFRA	Pinecone · DuckDB

#0759

TRADITIONAL

CUTTING-EDGE

Language Learning Emergent Behavior Testbed

Education

Push the frontier with a Emergent Behavior Testbed using LangGraph exploring novel methods for language learning — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Gemini 2.5 Pro
UI	React + Three.js + WebGL
BACKEND	Python · FastAPI + HPC scheduler integration
AI FRAMEWORKS	LangGraph OpenTelemetry PEFT / LoRA
VECTOR / INFRA	Weaviate · LanceDB

Predictive Maintenance Autonomous Research Lab

Manufacturing

Push the frontier with a Autonomous Research Lab using Ray exploring novel methods for predictive maintenance — reproducible benchmarks, ablation studies, failure-mode analysis and an open evaluation harness.

LLM	Custom fine-tuned 70B
UI	Custom React research console
BACKEND	Python · FastAPI + Ray Train + MLflow
AI FRAMEWORKS	Ray LangGraph NeMo Guardrails TRL
VECTOR / INFRA	Qdrant · OpenSearch

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